8. d) M_3N_2

if the total number of M atoms is n, then the number of tetrahedral voids =2n

given that $\left(\frac{1}{3}\right)^{rd}$ of tetrahedral voids

are occupied i.e., $\left(\frac{1}{3}\right)\times 2n$ are occupied by N atoms

$\therefore$ M:N $\Rightarrow n:\left(\frac{2}{3}\right)n$

$1:\left(\frac{2}{3}\right)$

$3:2 \Rightarrow M_3N_2$

9. c) 6

$$\frac{r_{C^+}}{r_{A^-}} = \frac{0.98\times10^{-10}}{1.81\times10^{-10}} = 0.54$$

it is in the range of 0.44–0.732, hence the coordination number of each ion is 6

10. d) $\left(\frac{\sqrt{3}}{2}\right)\times 400\text{pm}$

$\sqrt{3}\ a = r_{Cs^+} + 2r_{Cl^-} + r_{Cs^+}$

$\left(\frac{\sqrt{3}}{2}\right) a = \left(r_{Cs^+} + r_{Cl^-}\right)$

$\left(\frac{\sqrt{3}}{2}\right) 400 =$ inter ionic distance

11. a) $\left(\frac{100}{0.414}\right)$

for an fcc structure $\frac{r_{X^+}}{r_{y^-}} = 0.414$

given that $r_{X^+} = 100\,pm$

$r_{y^-} = \frac{100\,pm}{0.414}$

12. c) 32%

packing efficiency = 68%

therefore empty space percentage = (100-68) = 32%

13. b) 848.5pm

let edge length =a

$\sqrt{2}a = 4r$

$a = \frac{4\times\ 300}{\sqrt{2}}$

$a = 600\times1.414$

$a = 848.4$ pm

14. b) $\left(\frac{\pi}{6}\right)$

$$\left(\frac{\frac{4}{3}\pi r^3}{a^3}\right) = \left(\frac{\frac{4}{3}\pi\left(\frac{a}{2}\right)^3}{a^3}\right) = \left(\frac{\pi}{6}\right)$$

15. a) excitation of electrons in F centers

16. c) $\left(\frac{1}{2}a:\frac{\sqrt{3}}{4}a:\frac{1}{2\sqrt{2}}a\right)$

$sc \Rightarrow 2r = a \Rightarrow r = \frac{a}{2}$

$bcc \Rightarrow 4r = \sqrt{3}a \Rightarrow r = \frac{\sqrt{3}a}{4}$

$fcc \Rightarrow 4r = \sqrt{2}a \Rightarrow r = \frac{\sqrt{2}a}{4} = \frac{a}{2\sqrt{2}}$

$\left(\frac{a}{2}\right):\left(\frac{\sqrt{3}a}{4}\right):\left(\frac{a}{2\sqrt{2}}\right)$

17. d) $\left(\frac{\sqrt{3}}{2}\right)a$

if a is the length of the side, then the length of the leading diagonal passing through the body centered atom is $\sqrt{3}a$

Required distance $= \left(\frac{\sqrt{3}}{2}\right)a$

18. a) 915 kg m-3

$\rho = \frac{n \times M}{a^3 N_A}$

for bcc

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